university of groningen
faculty of science and engineering
bernoulli institute for mathematics, computer science and artificial intelligence

Distributed Systems Group

Andrés Tello

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university of groningen / faculty of science and engineering / bernoulli institute for mathematics, computer science and artificial intelligence



Area(s) of expertise

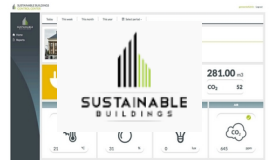
- Big Data and Cloud Computing
- Pervasive Computing and Smart Environments
- Smart Industry and Internet of Things
- Digital Twin for Smart Systems and Environments
- Adaptive / Scalable / Sustainable Infrastructure for Distributed AI
- Applied AI (water, energy, smart buildings, road traffic, manufacturing, data centers)

Challenges

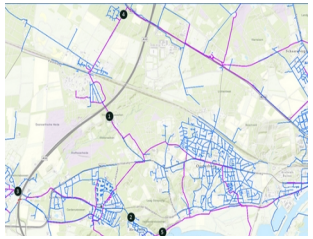
Adaptive and Scalable and Sustainable and Responsible Software

- automated composition, orchestration and recommendation of data and ML-pipelines
- AI as a product: removing a gap between AI/data scientist's laptop and production
- computation-centric modeling of complex systems (e.g., GNNs)

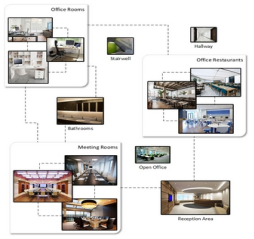
keywords: big data, data science and data engineering, cloud services, machine learning



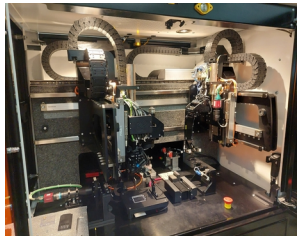




Water



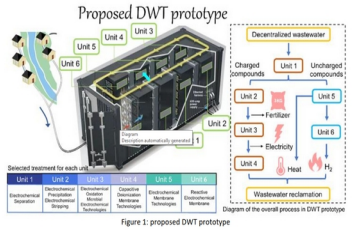
Energy



Smart Manufacturing



Road Traffic



Digital Twins

RESEARCHABLE

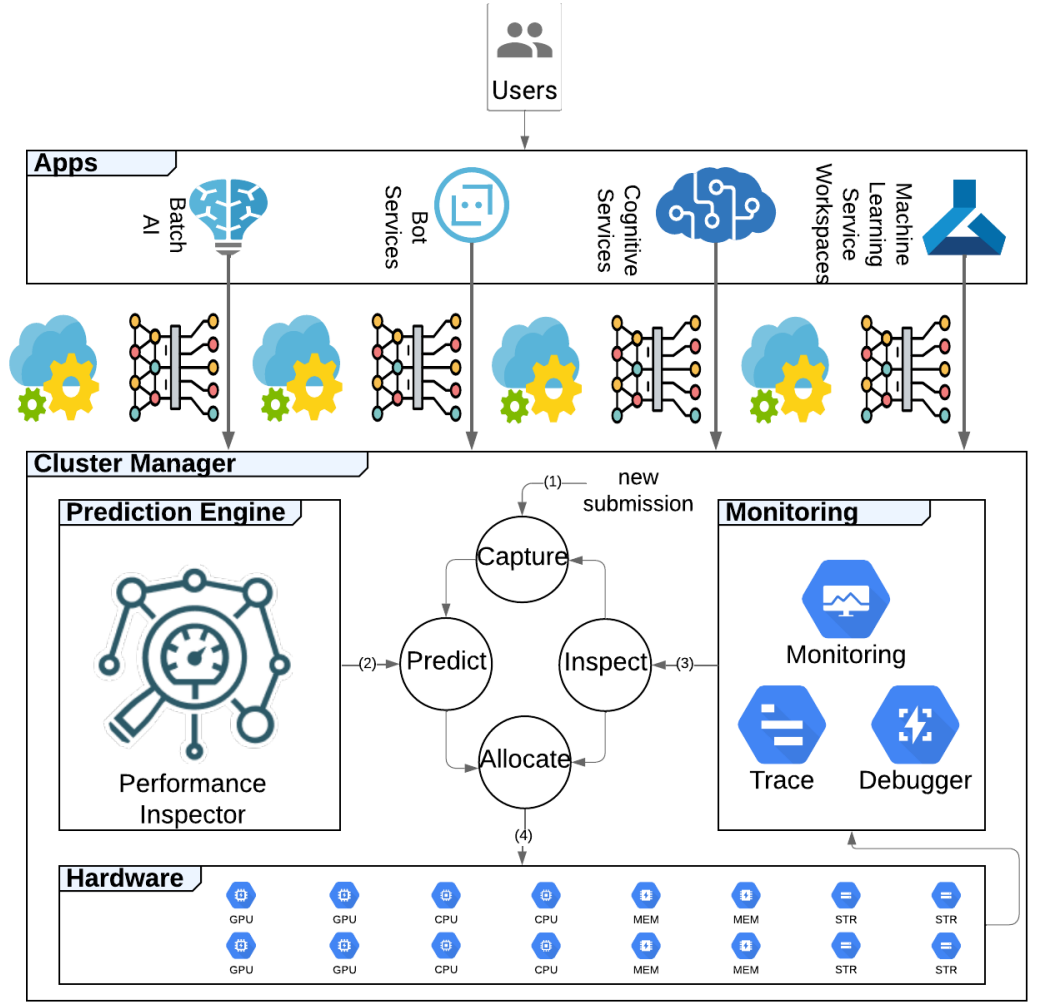
CAIDEL
Continuous AI Delivery

Teaching

Course Title	Level	Coordinator
Advanced Programming	Bachelor CS	H. Groefsema
Bachelor's project	Bachelor CS	D. Düstegör (coordinator)
Data Challenges in AI Systems	Master CS	D. Düstegör (coordinator)
Fundamentals of Distributed Systems	Bachelor CS	D. Düstegör (coordinator)
Research Skills in Computing Science	Bachelor CS	D. Düstegör (coordinator)
Scalable Computing	Master CS	A. Lazovik (coordinator)

Cloud Computing - System Design for AI

- Developing a prediction engine for estimating the performance of DNN models.
- Optimizing the system's performance of ML workload running on GPU cluster.
- Scheduling and resource allocation of AI workloads on GPU Cluster.
- Developing monitoring and profiling tools for collecting jobs and hardware statistics.



Contact
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Automated compliance verification

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Introduction

The increased use of digital technologies by organisations sparks a rise in ***legislation***. European laws and directives such as the AI act demand ***regulatory compliance systems***.

Implementing regulatory compliance is a key and pressing challenge for organisations to ***ease audits***, and for governments to ***protect citizen rights***.

Problem

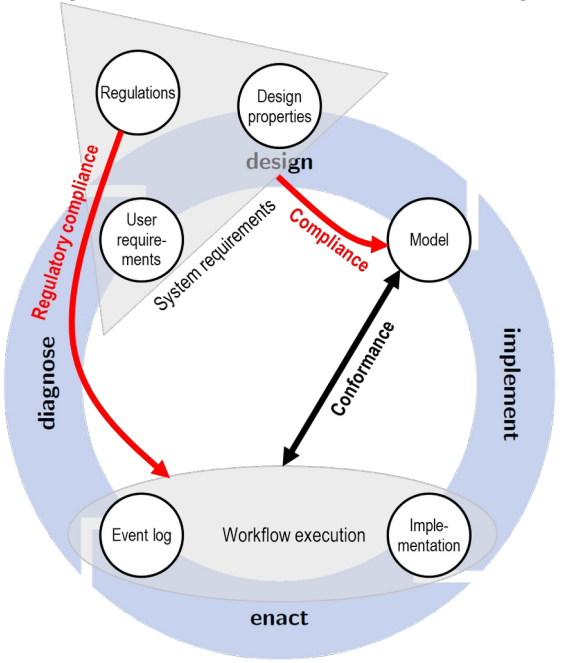
The problem of regulatory compliance verification quickly becomes undecidable. As a result, we must design ***smart practical applicable systems***.

Student projects & challenges

- Verifying ***Environmental Sustainability***
- Verifying ***Security*** and ***Privacy*** concepts
- ***Predicting*** violations
- ***Avoiding*** violations
- Obtaining ***regulations from laws***
- Obtaining ***regulations from event logs***
- ***Sharing*** regulations
- ***Aligning events*** against advanced graph models
- Automated ***creation of compliant models***

Compliance in the process life cycle

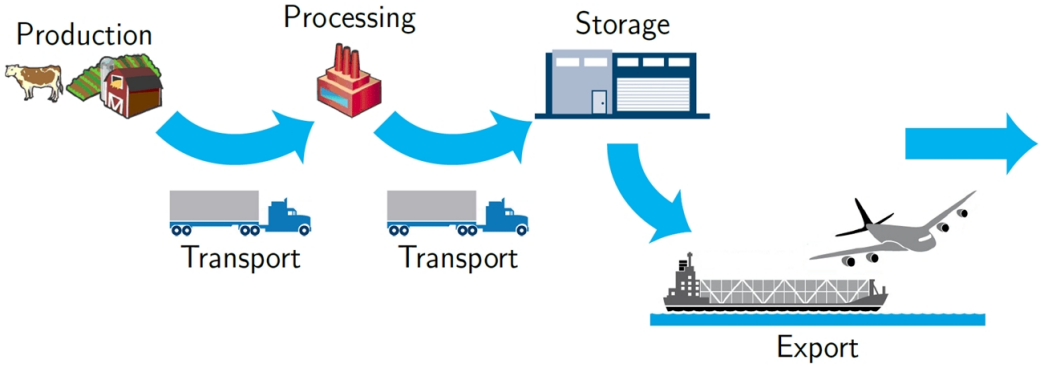
Compliance can be ***formally*** analyzed at different stages of the process life cycle. At design time we can use ***model checking*** to see if the designed model follows the system requirements and during/after runtime we can use ***runtime verification*** to monitor obligations to see if recorded events follow regulations.



Patented approach

"log data compliance", Governatori, G., van Beest, N., Cryer, A. & Groefsema, H. (Inventors), 16-Dec-2021, Patent No. WO2021248201, 10-Jun-2021, Priority date 11-Jun-2020

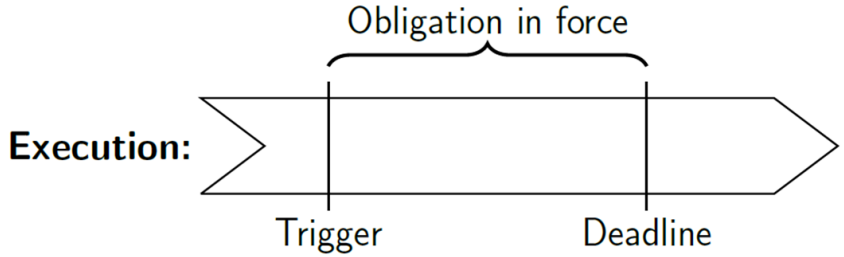
Cotton harvest



Regulation: *"Seeds must be separated from the crop before export."*

Solution: Use ***runtime verification*** to track transports and ***monitor*** crop weights before and after processing. Crop weights must be a specific percentage before and after processing.

Formalizing rules as obligations



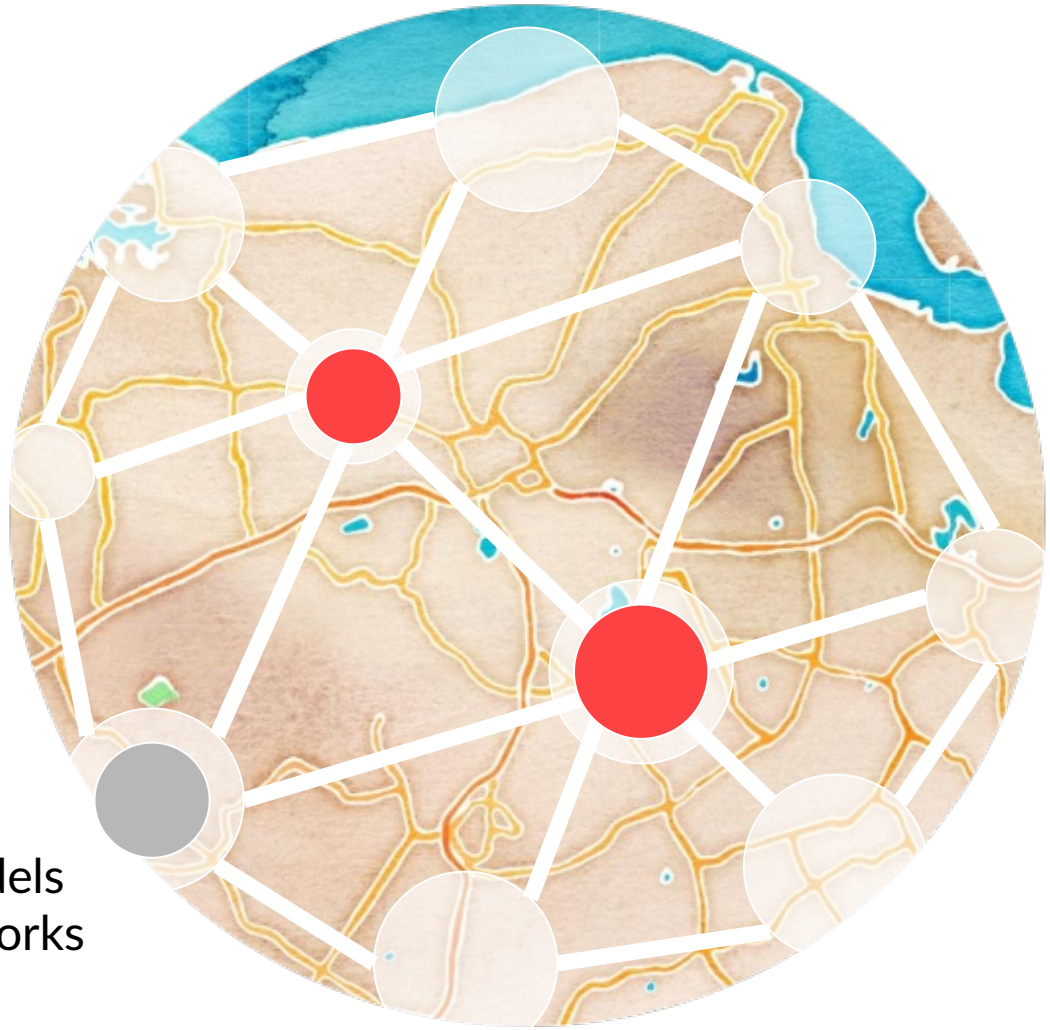
An ***obligation*** $\mathcal{O} = \langle \mathcal{Z}, t, \varrho, \tau, \delta \rangle$ is a 5-tuple with identifier function $\mathcal{Z}$, the type t of obligation being an achievement or maintenance obligation, and requirement ϱ, trigger τ, and deadline δ being logic formulas over data contained within events. Once the trigger τ holds the requirement ϱ must hold until δ holds for some future event.

DiTEC: Graph Neural Networks for Digital Twin on Water Distribution Systems

Problems

> Monitoring Water Systems

> Detect Out-of-Distribution
> Compromise solutions



Methods

> Graph Neural Networks

> Self-Supervised Learning

> Mixture of Experts

Interest area

> Simulation Incorporation

> Multimodal Foundation Models
> Bio-plausibility Neural Networks

Student projects

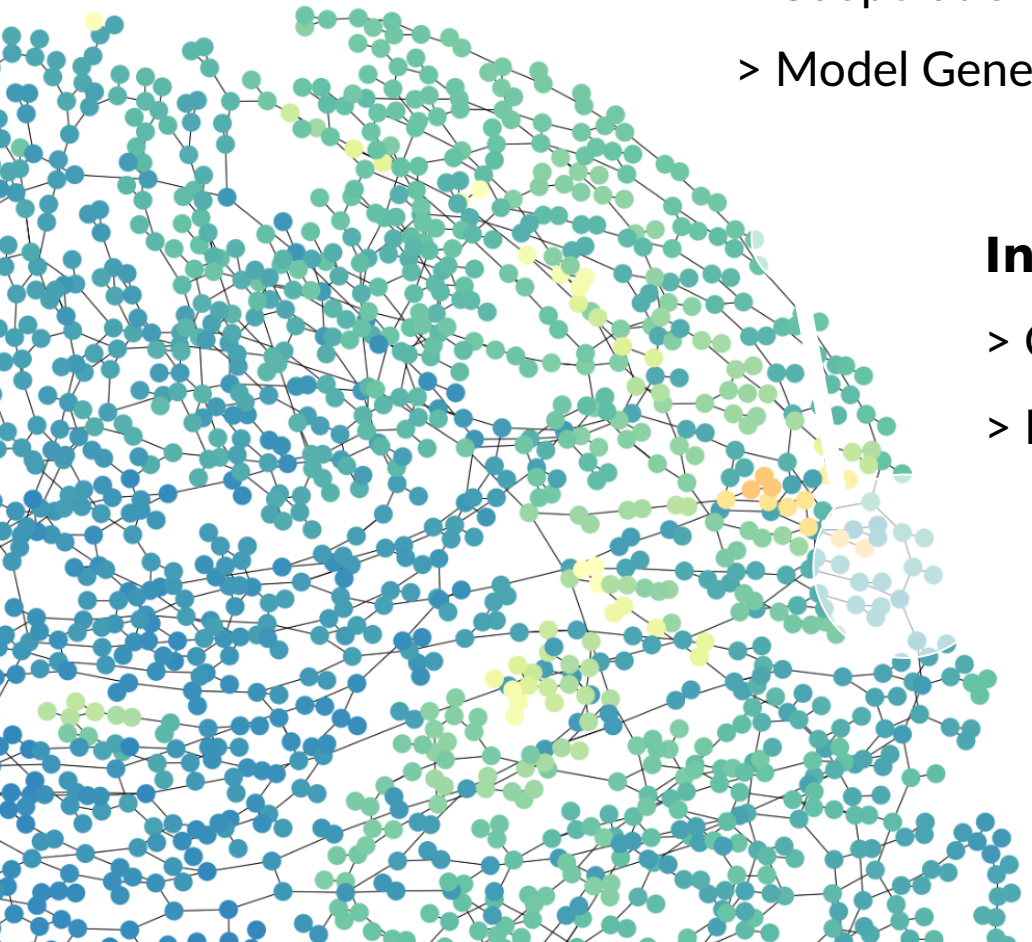
> State Estimation

> Demand Forecasting

> Surrogate Modeling

Huy Truong | h.c.truong@rug.nl

GLOWED: Global-Local Operations on Water by Evolutionary Digitalization



Problems

> Cooperation between Distributed DTs

> Model Generalization

Methods

> Graph Representation Learning

> Self-Supervised Learning

Interest area

> Graph Foundation Models

> Federated Learning

Student projects

> Leakage Detection Algorithms

> Energy optimization

> Pre-training strategies for GraphML

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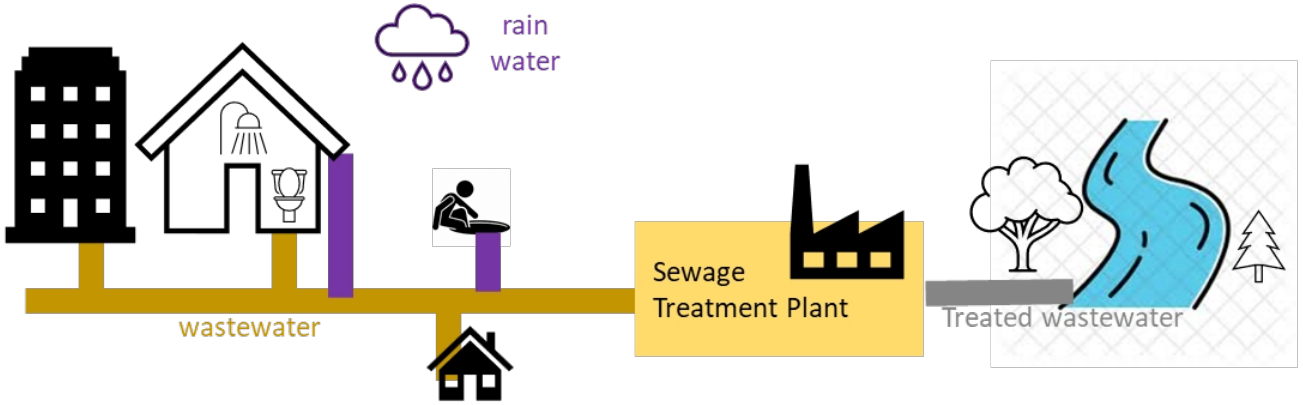
DDTClean: Distributed Digital Twin for Clean Water

Problems

> High water level in streets during heavy rain

> Overflow in the city (e.g. from manholes)

> Decreasing water purity (e.g. in canals)



Technical:

> Hydraulic models need calibration after heavy rain

> Limited sensor coverage (water level in few manholes, groundwater level sensors)

Objectives

> Monitoring wastewater systems (sewage system, treatment plan)

> Control storm overflows (relief valves) to redirect excess into canals

Methods

> Data driven methods (Deep Learning, Graph Neural Networks)

Student projects

> State Estimation

> Optimal Sensor Placement

> Overflow Forecasting

> Surrogate Modeling

Dilek Düştegör & Revin Alief
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Attention-based GNNs for Water Distribution Networks

- Context
- Problem
- Graph-based models for WDNs
- Conclusions

Context

Monitoring the state of a water distribution network

Improve management
Reduce operations costs
Leak localization

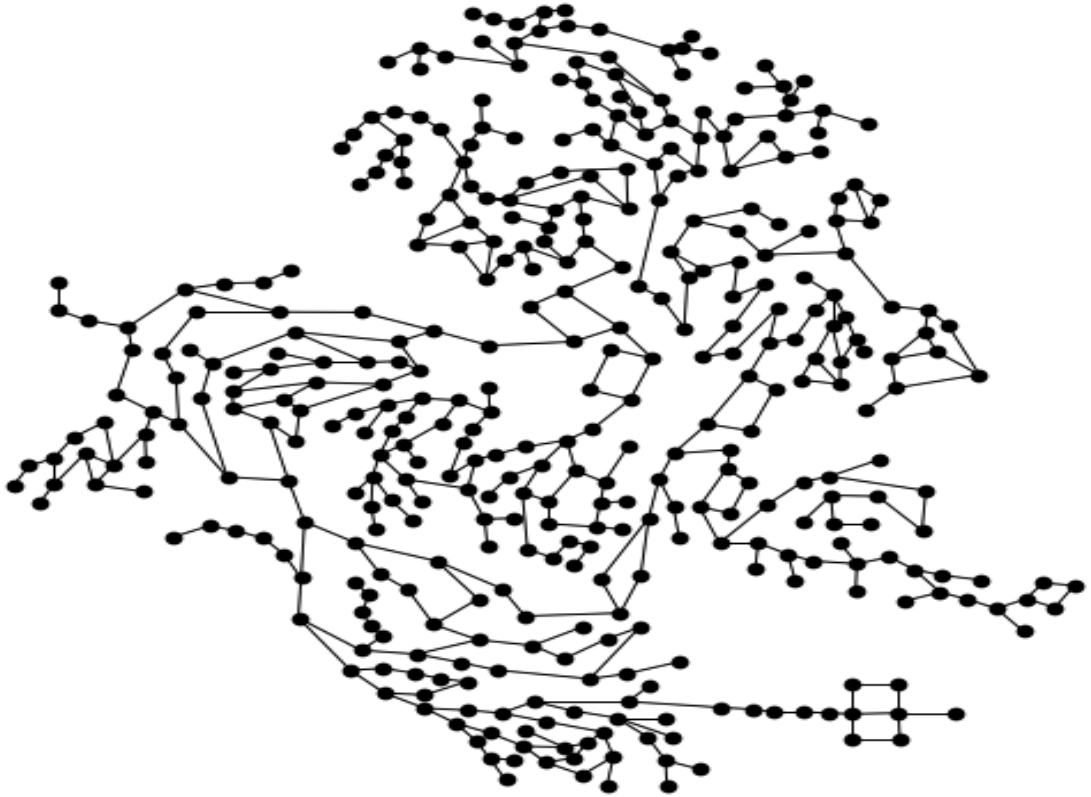


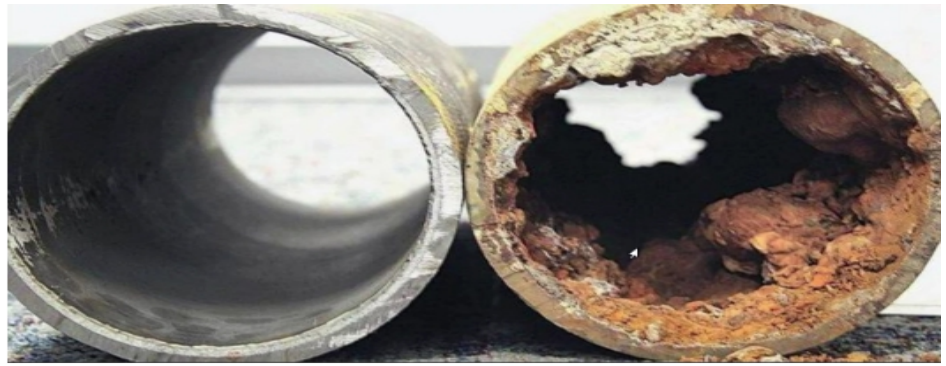
Image generated from C-town WDN dataset (Riccardo et.al, 2018)



Non trivial

Uncertainty and lack of reliable data











Context

Monitoring the state of a water distribution network

Install sensors

Gather data

Data Analysis

Expected Sensor Coverage

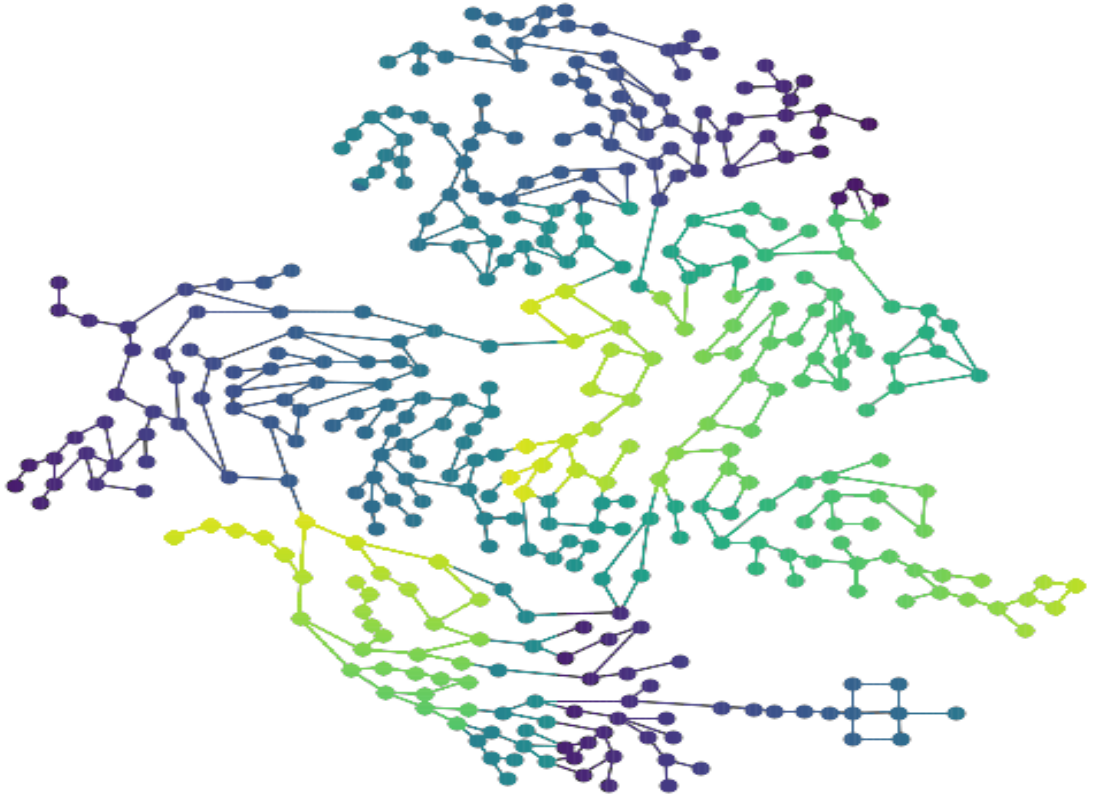


Image generated from C-town WDN dataset (Riccardo et.al, 2018)

Context

Monitoring the state of a water distribution network

high installation & maintenance costs

Actual Sensor Coverage

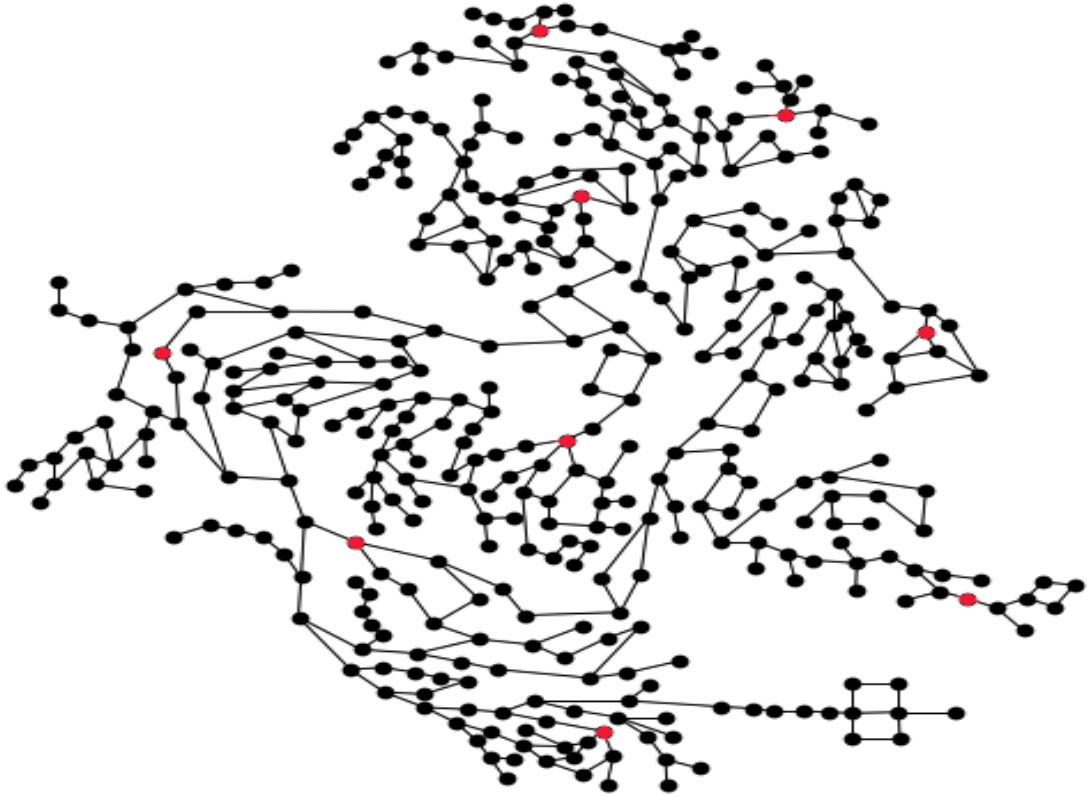
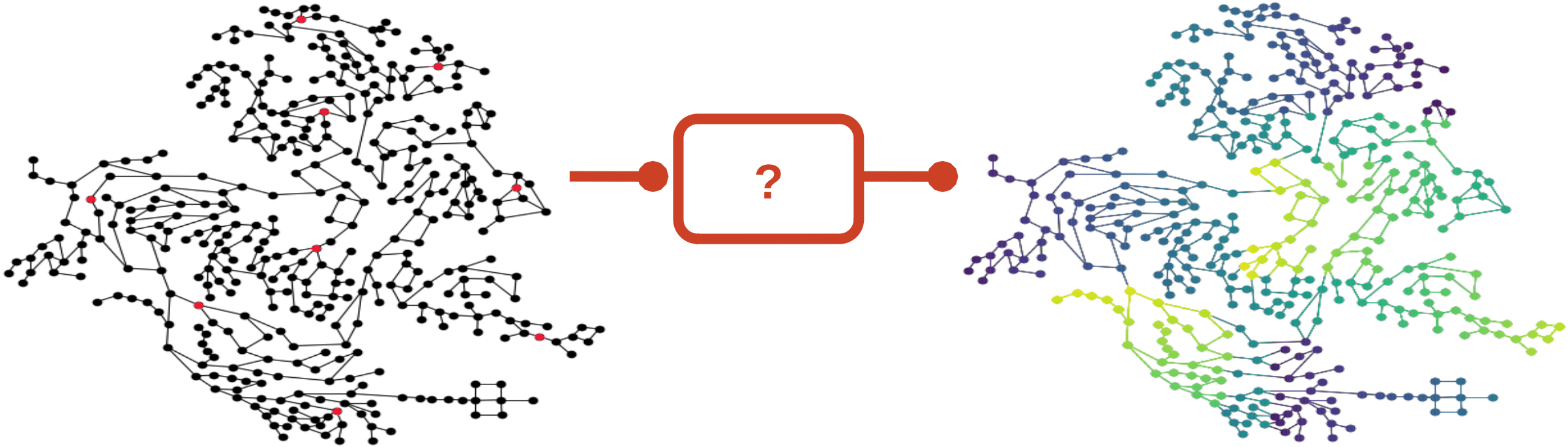


Image generated from C-town WDN dataset (Riccardo et.al, 2018)

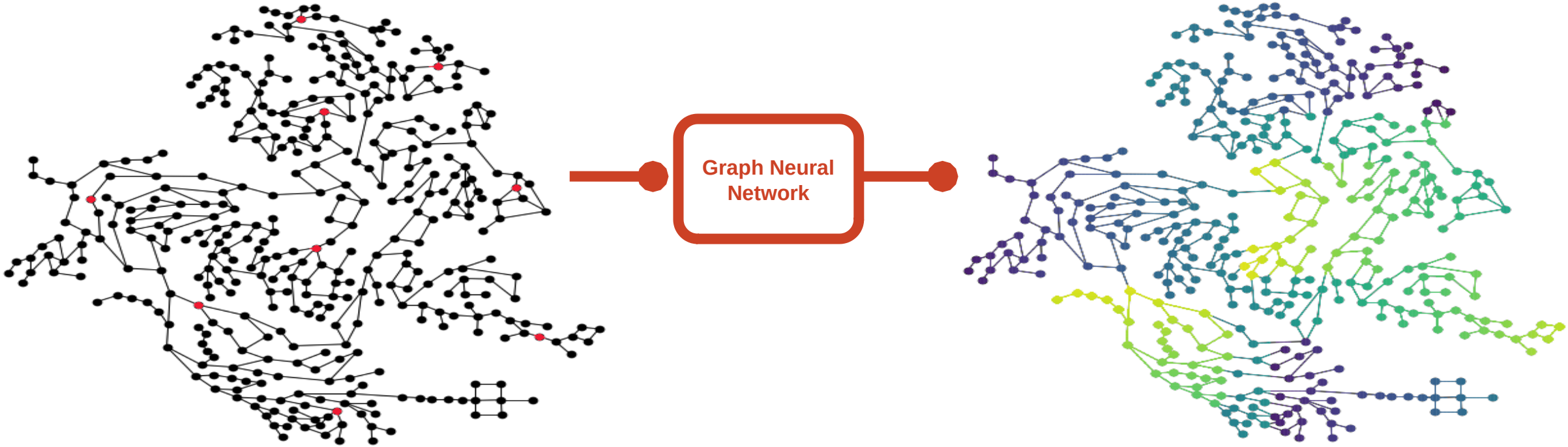
Problem

Reconstruct the pressures of the entire water distribution network using few existing sensors



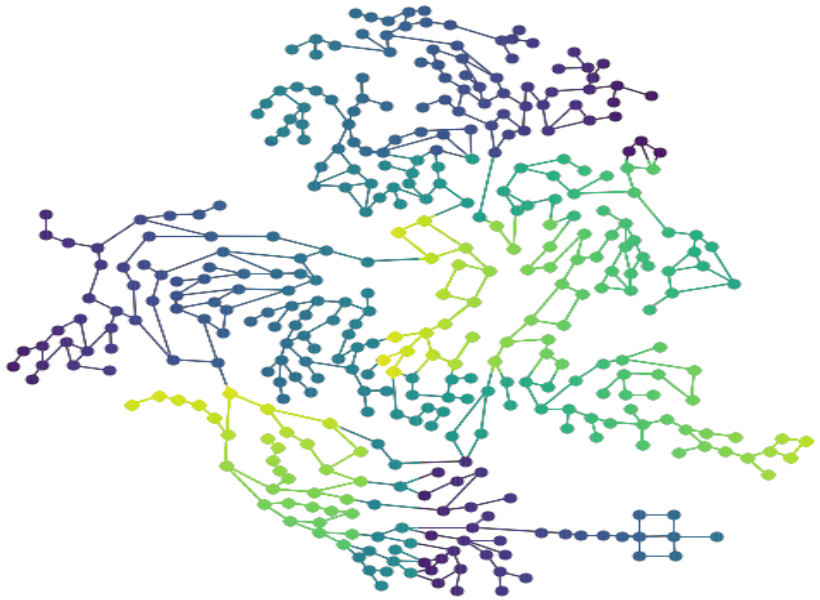
Our approach

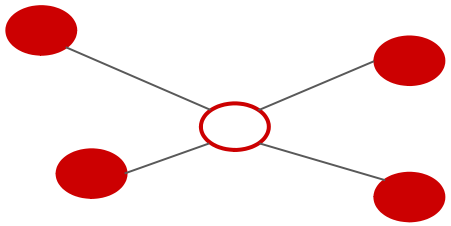
Reconstruct the pressures of the entire water distribution network using few existing sensors



Graph Neural Network models (GNNs)

How to define scopes and directions in graph?

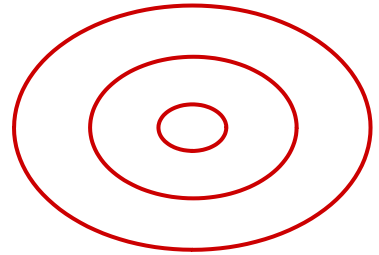




Spatial Domain

GAT (Veličković et. al, 2018)

GCN (Kipf & Welling, 2017)



Spectral Domain

Chebnet (Defferrard, 2016)

Cayleynets (Levie et. al, 2018)

2017

Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.0 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature.

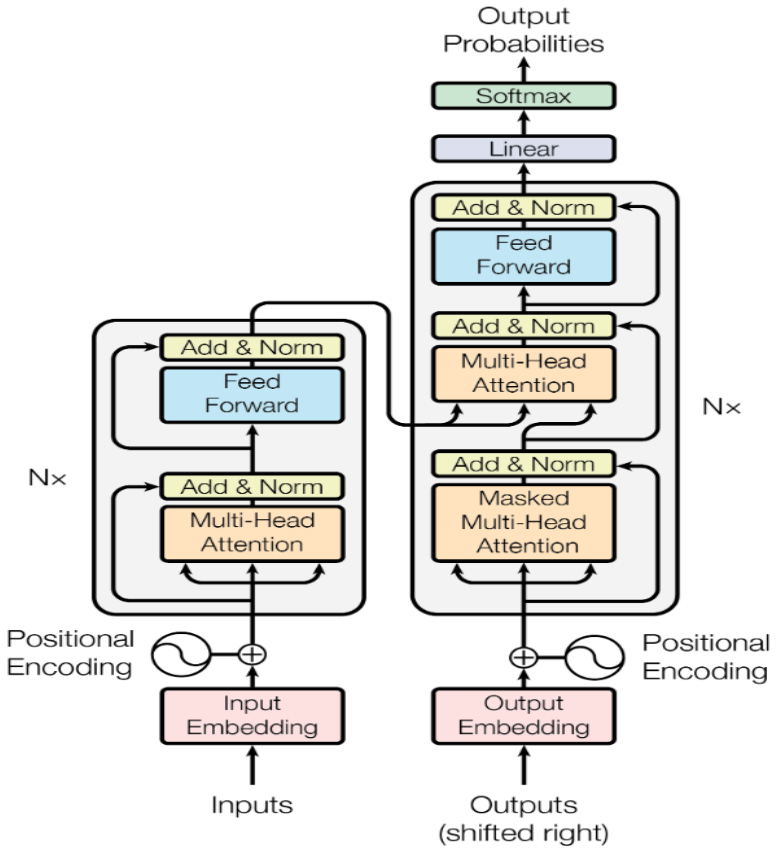
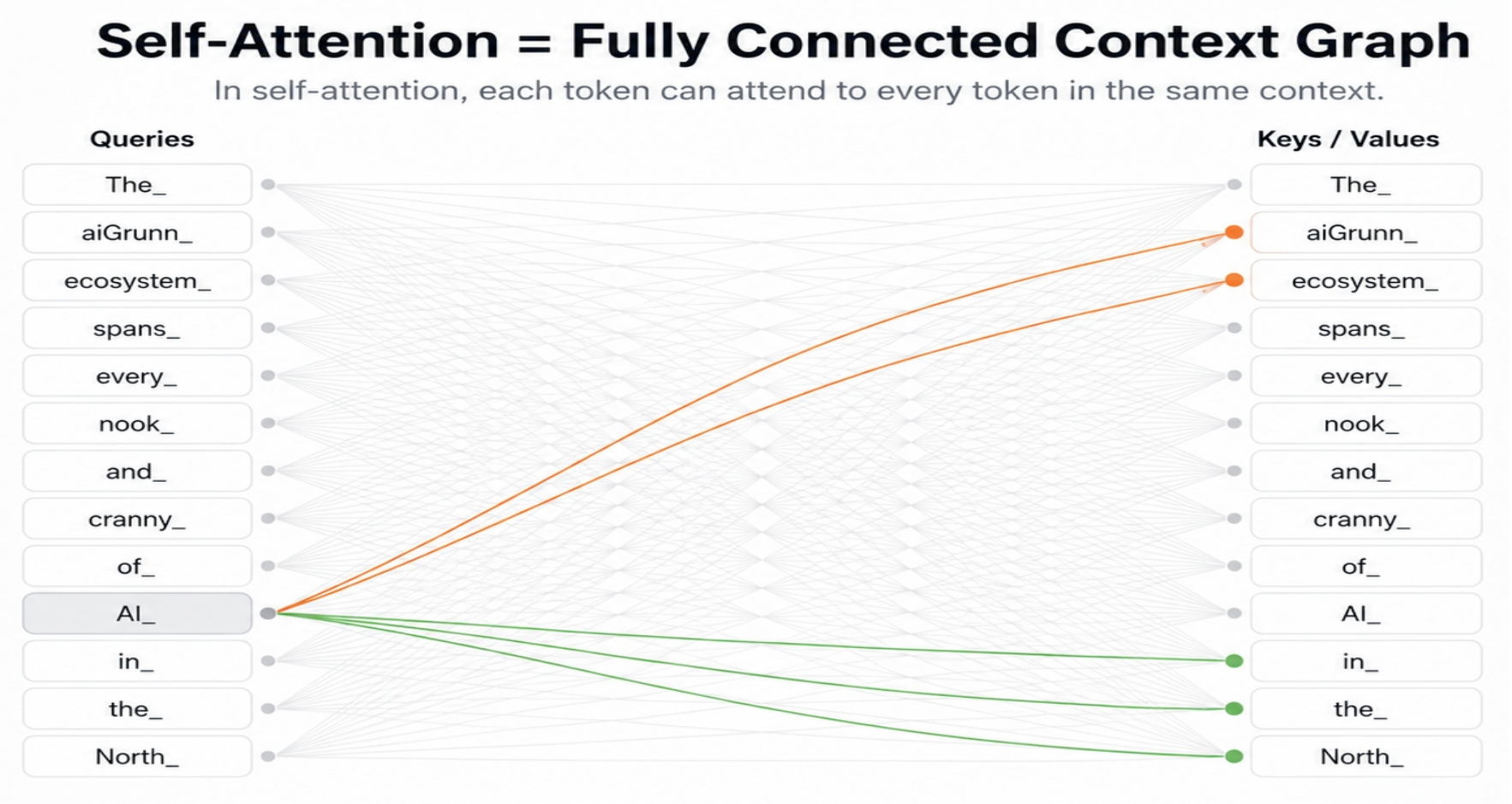


Figure 1: The Transformer - model architecture.

Post-Transformer AI Models Landscape

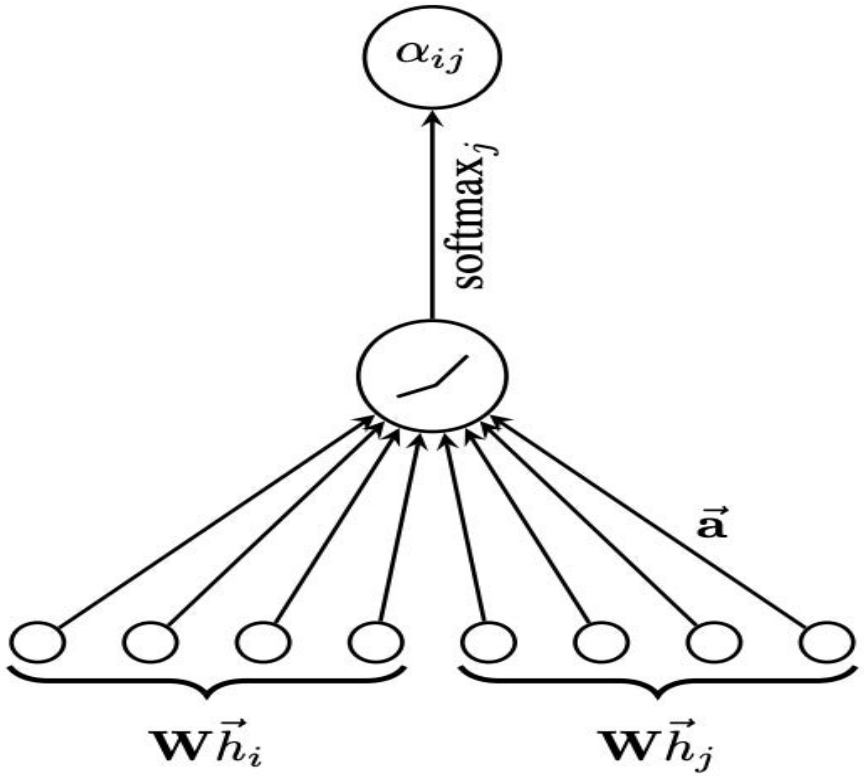


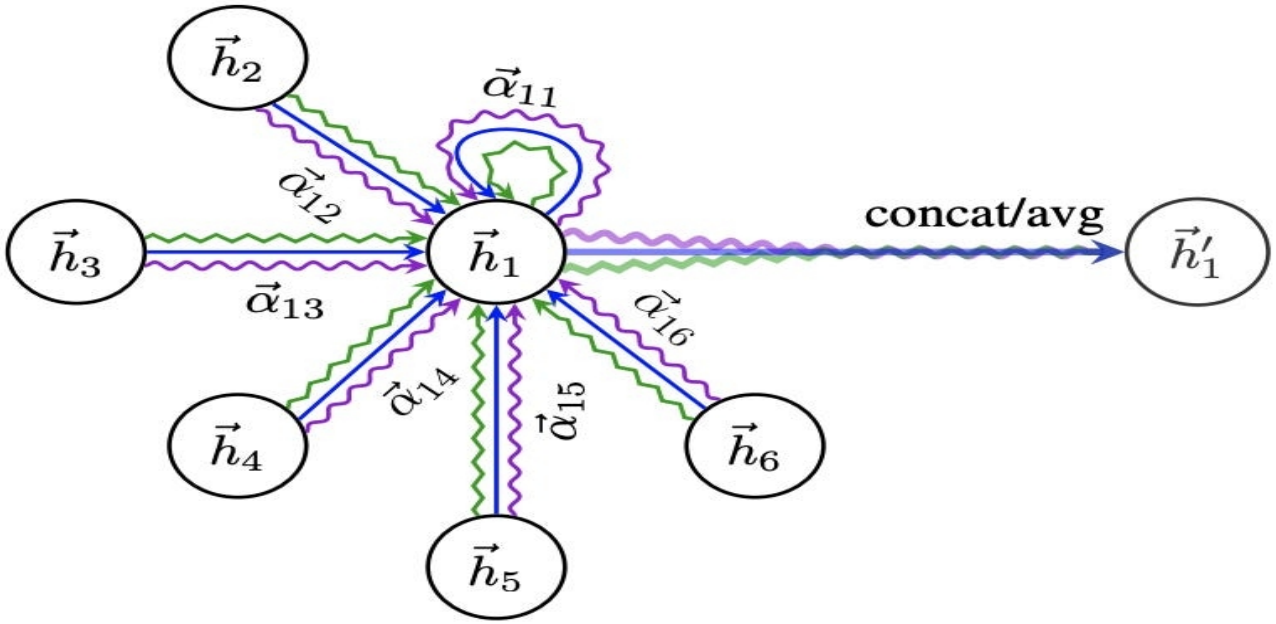
Transformer are GNNs





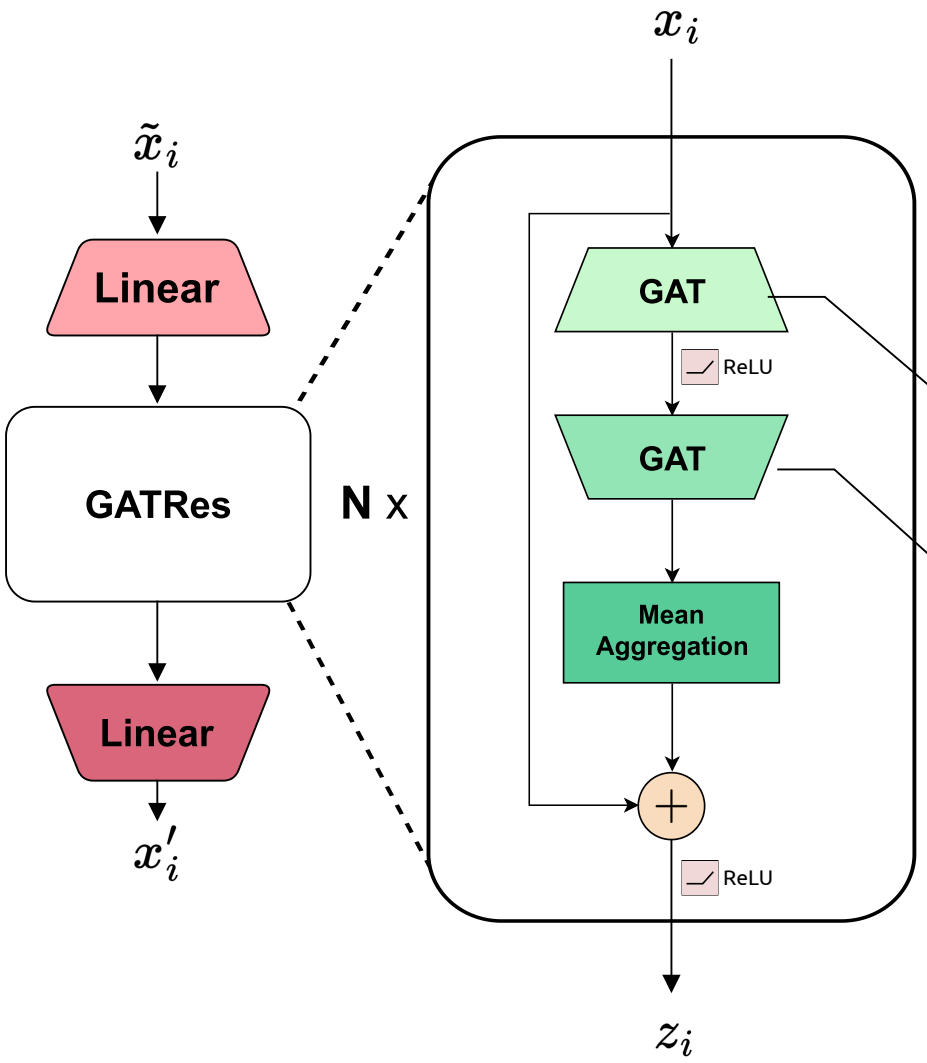
GNN Attention



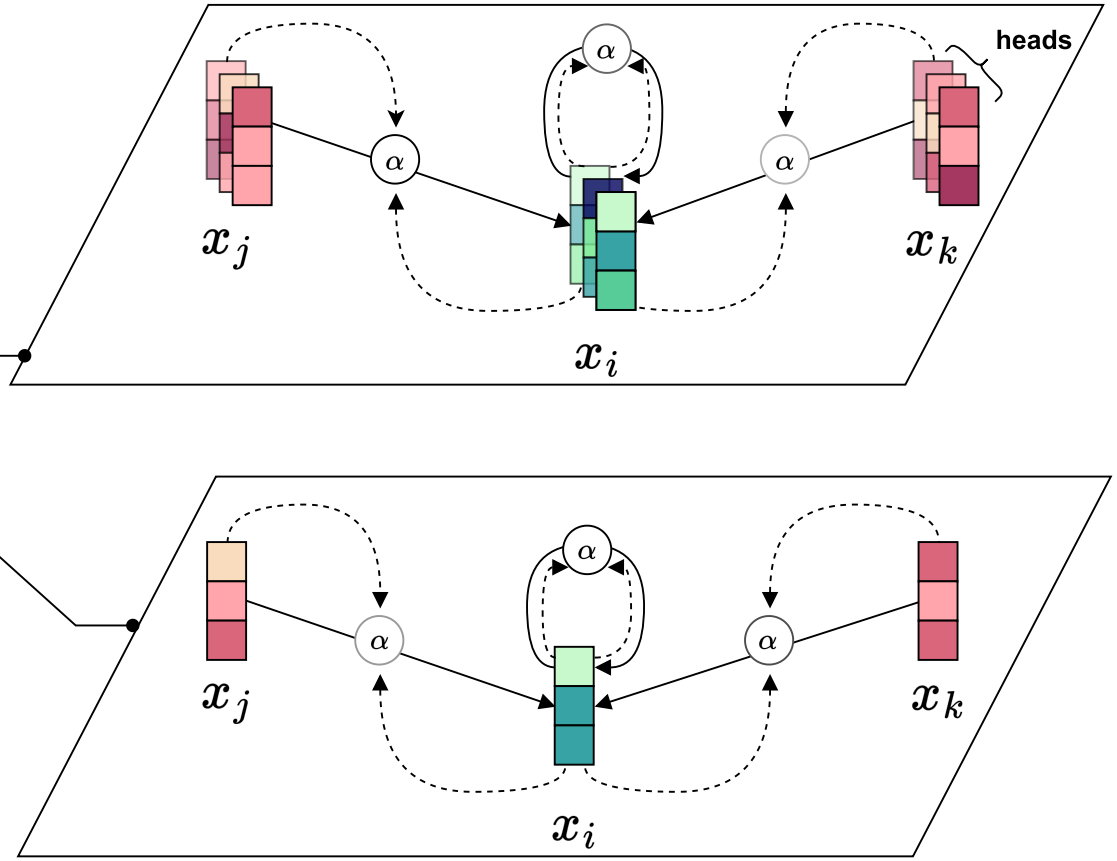


Graph Attention Networks. Velickovic et al., 2018

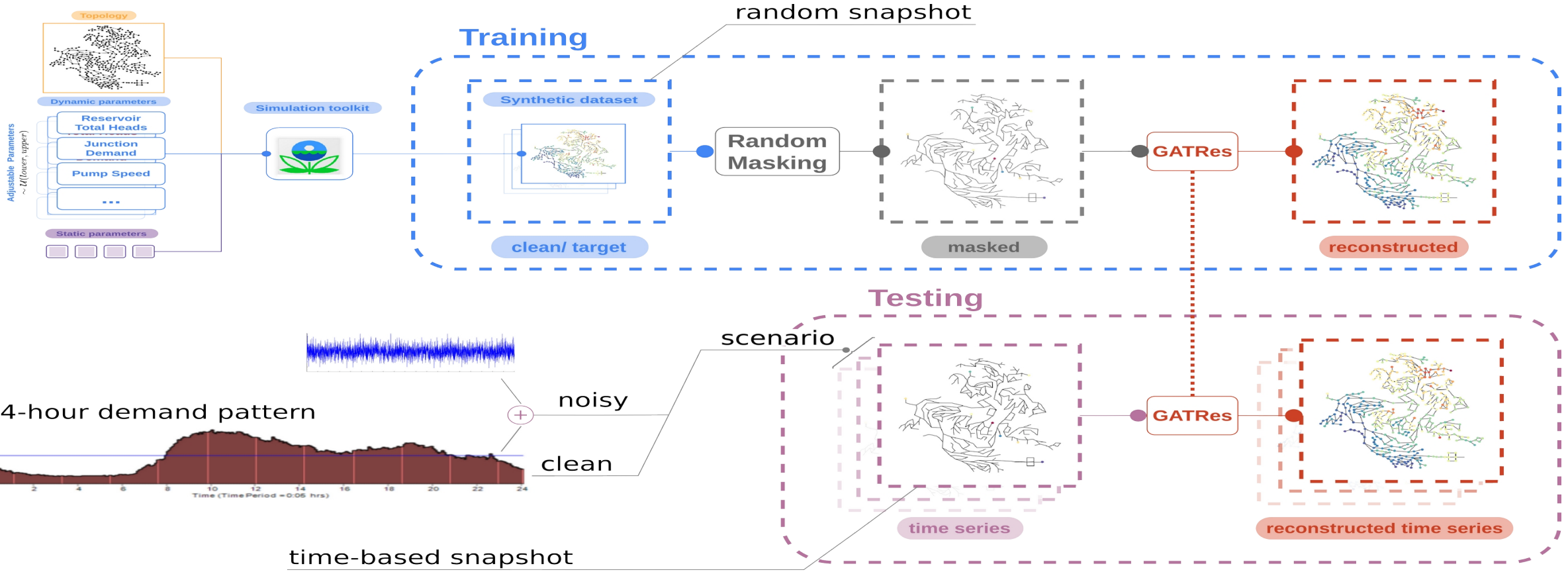
GATRes Model



GAT with Residual Connections

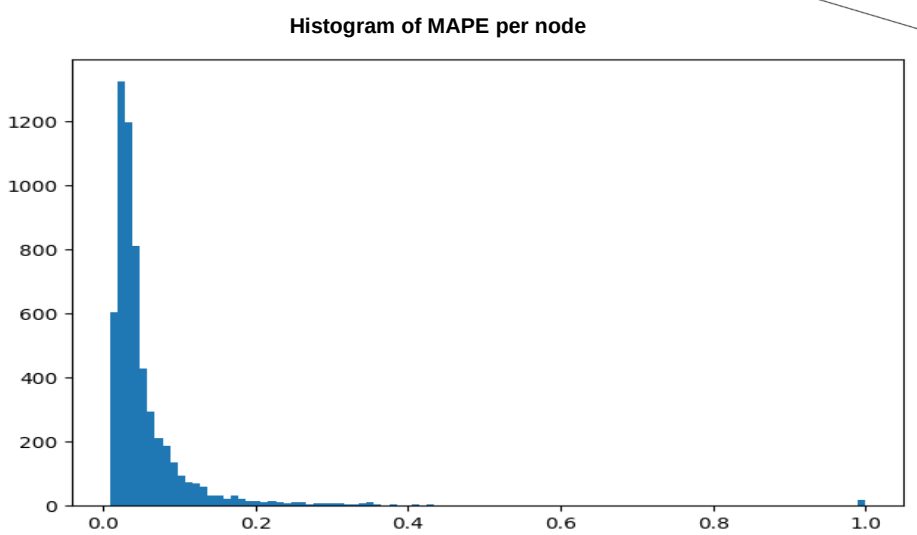


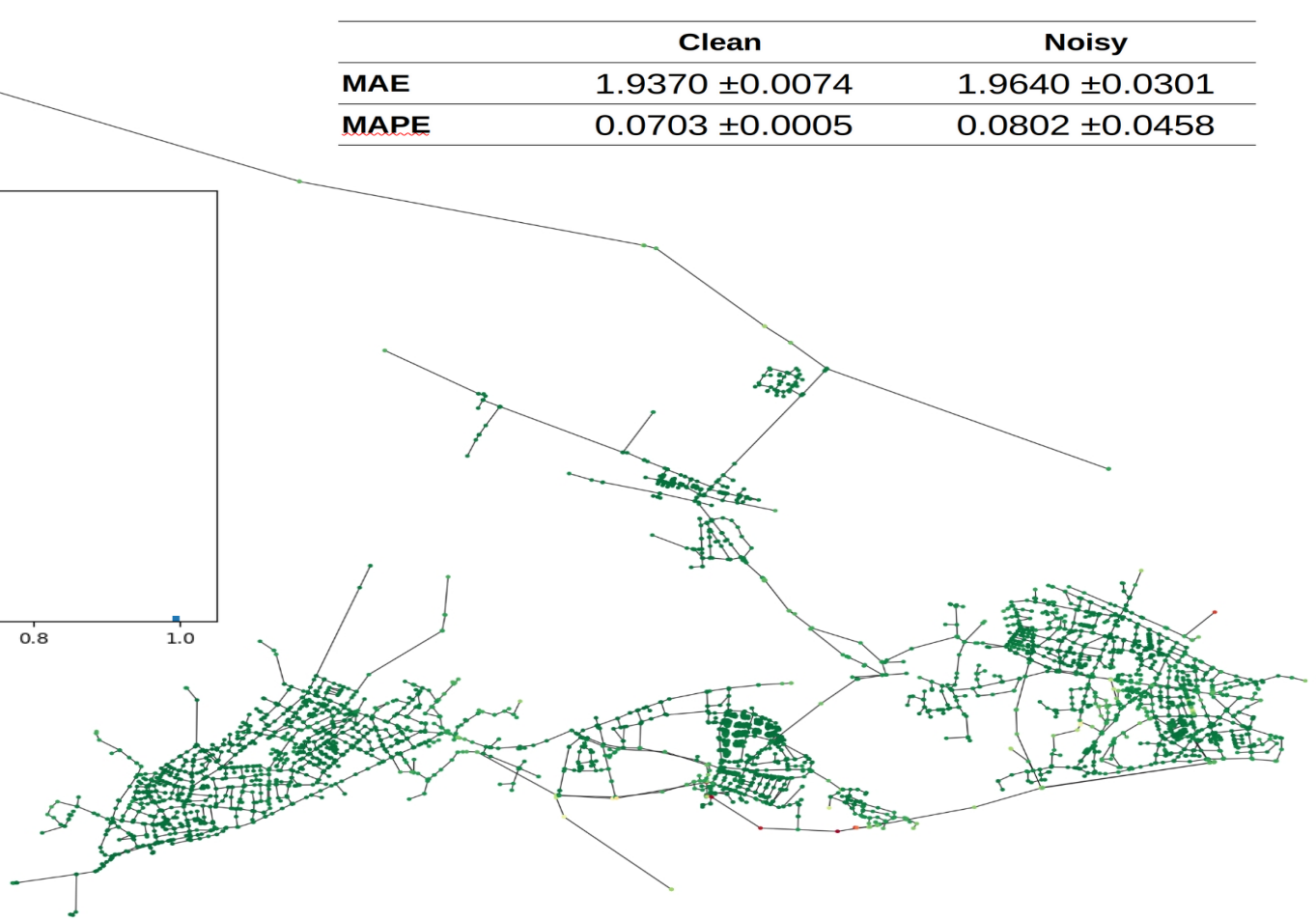
Model Training & Evaluation



Single Model Performance

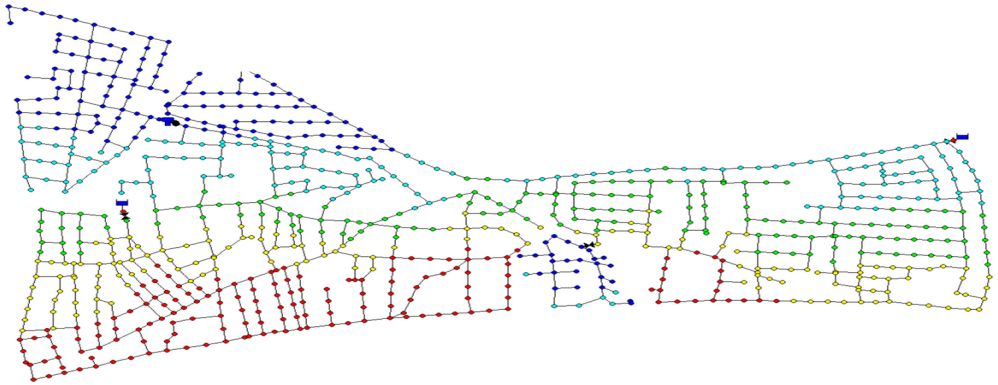
	Clean	Noisy
MAE	1.9370 ±0.0074	1.9640 ±0.0301
MAPE	0.0703 ±0.0005	0.0802 ±0.0458

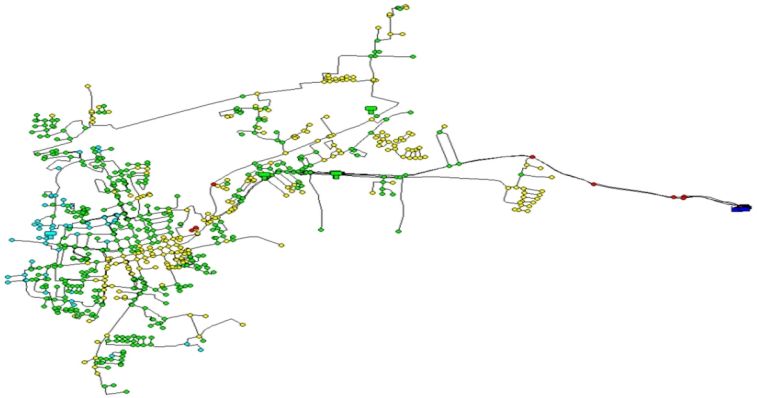


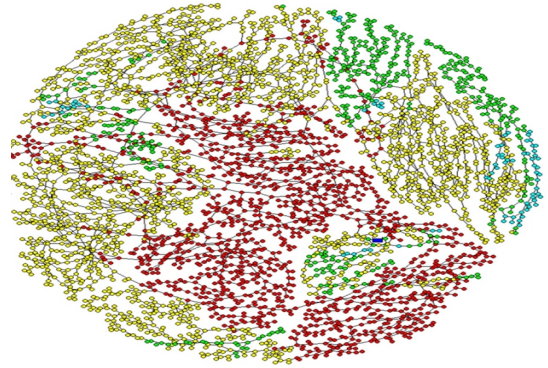


GATRes Generalization

model training on multiple WDN





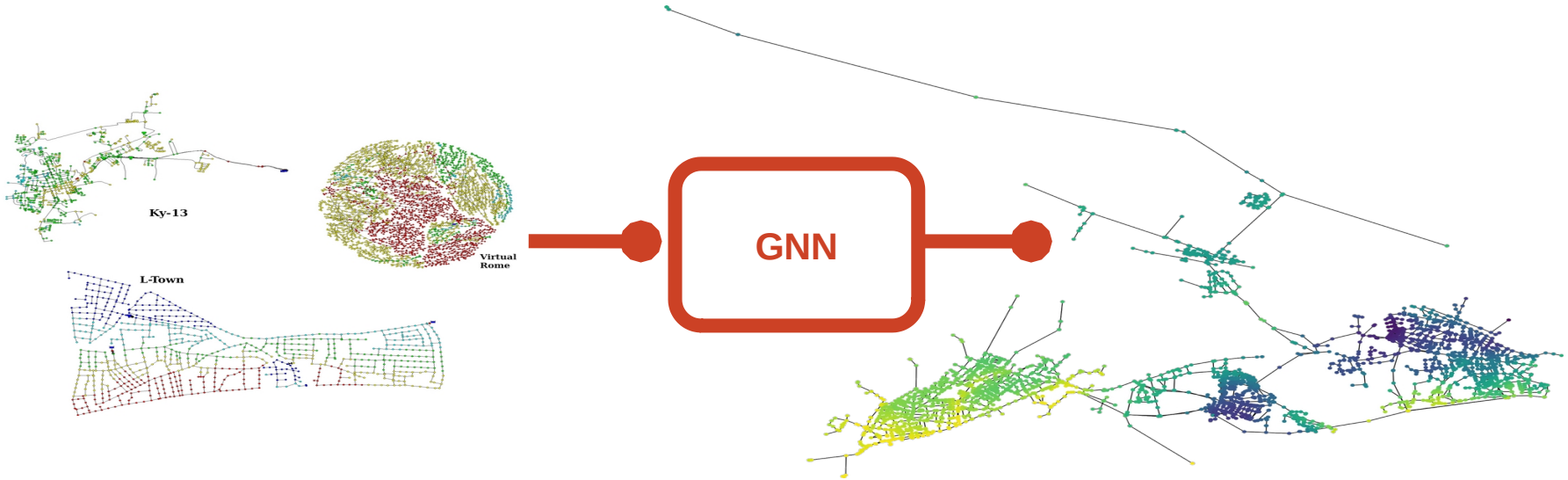


	L-Town
	Vrachimis et al. 2022 (BattLeDIM)
nodes	782
edges	905

Ky-13
University of Kentucky
770
915

Large
University of Innsbruck
3557
6170

GATRes Generalization



Models	MAE (↓)	MAPE (↓)
Single-Graph	1.9370 ±0.0074	0.0703 ±0.0005
Multi-Graph Zero-Shot	3.0597 ±0.0074	0.0998 ±0.0005
Fine-tuned	**1.9097** ±0.0076	**0.0695** ±0.0005

How to improve Generalization and Transferability in WDNs domain?

- Deep Learning has contributed to significant advances
  - Computer vision, Natural Language Processing.
  - **Water Management** is not an exception

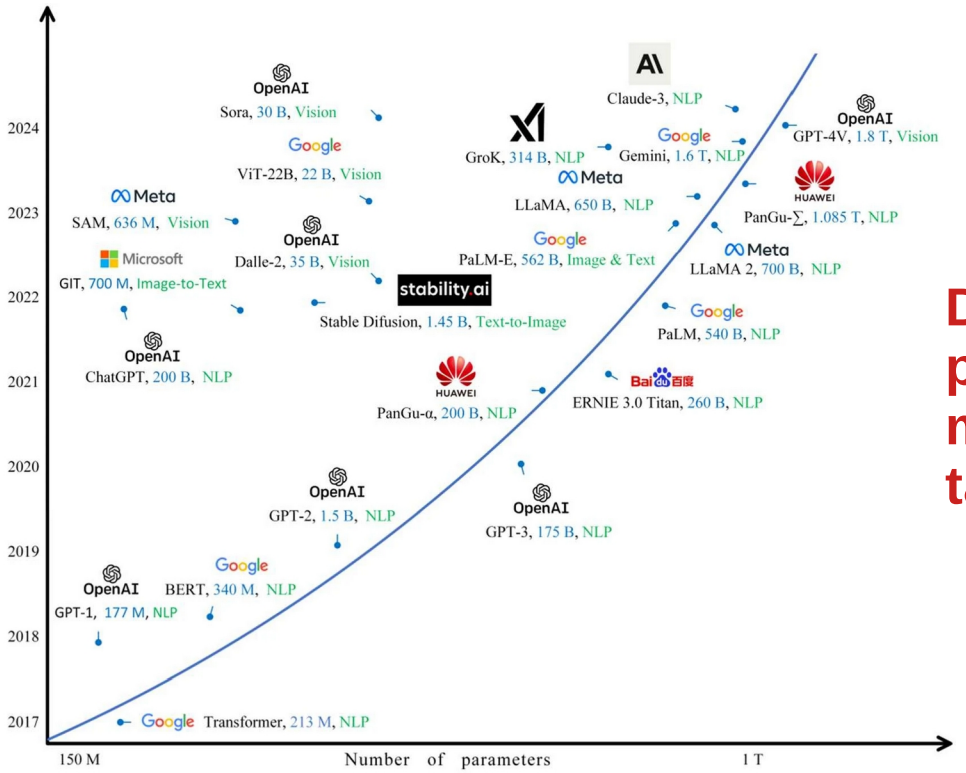
- Technological advances
  - Large-scale: computation, data and models
  - **Foundation Models**.

Foundation Model

"A **foundation model** is any model that is trained on broad data (generally using self-supervision at scale) that can be adapted (e.g., fine-tuned) to a wide range of downstream tasks"

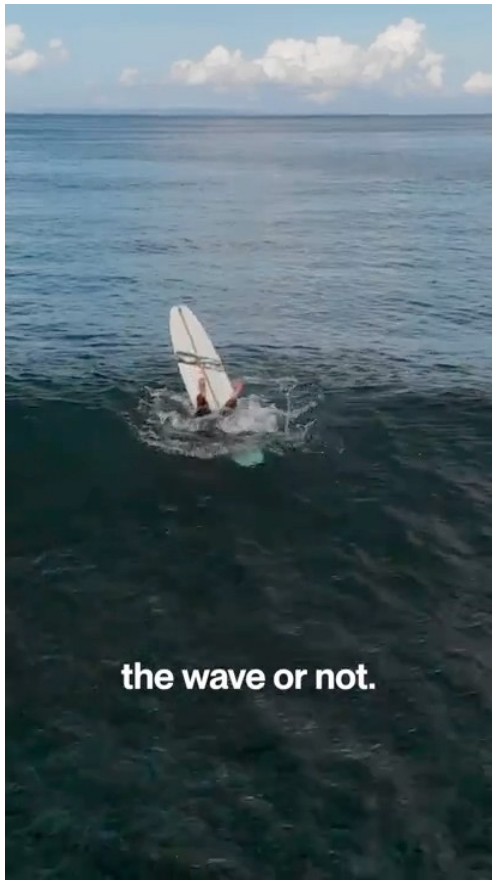
(Bommasani et al., 2021)

Foundation Models



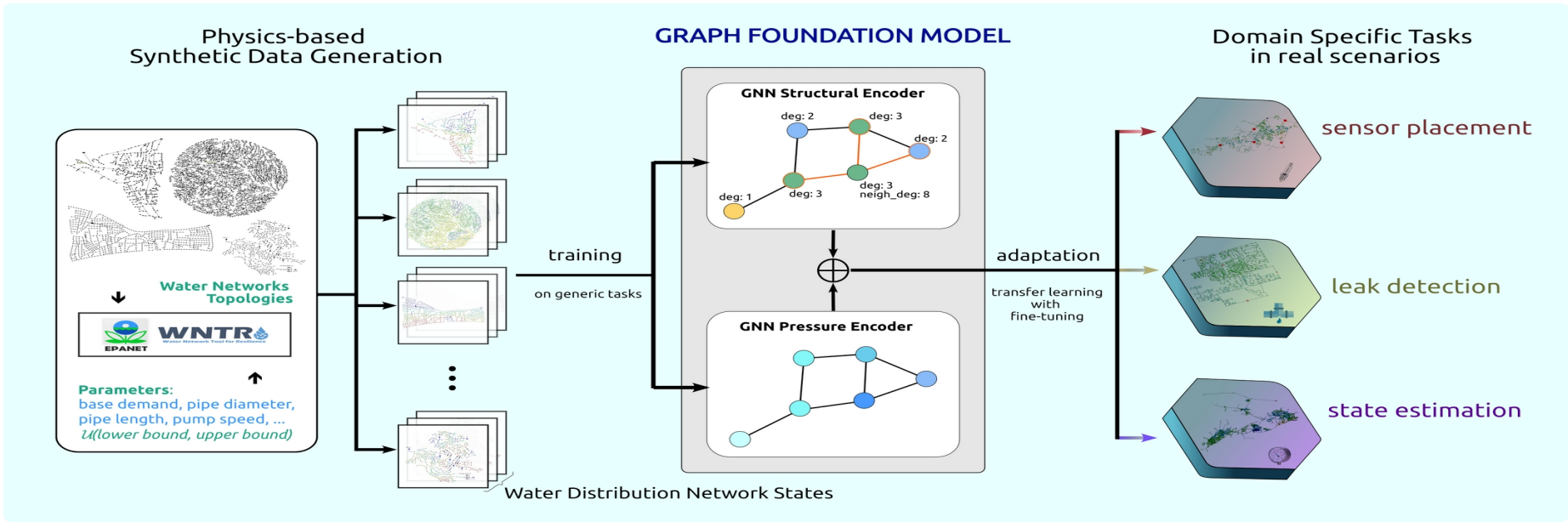
Tu, et al., An overview of large AI models and their applications, 2024

Do we really need billion-parameter general-purpose models for every domain or task?



FMs design/use according the constraints of each domain.

Graph Foundation Models for Water Distribution Networks



OUTCOMES

- Generalization capabilities
- Model reusability
- Faster deployment of applications
- Openly accesible training data

Overview of our proposal

Tailoring GNNs training to WDNs

- GNNs and Water domain
  - Centrality metrics: betweenness, closeness
    - Shortest paths are relevant to characterize the behavior of spatial networks
    - Edge-centrality

[Sitzenfrei et al., 2020; Giudicianni et al., 2021]

  - Node's degree
    - (Neighborhood) Node degree distributions for WDNs classification.

[Giustolisi et al., 2017]

    - Generalization capabilities of GNNs

[Yehudai et al., 2021; Zhu et al., 2021; Lee and Yoon, 2022]



DiTEC WDNs Dataset

Hugging Face

- **36 WDNs Baselines**
  - small (19 - ~500 nodes),
  - medium (500 - ~1000 nodes)
  - large (> 1000 nodes)

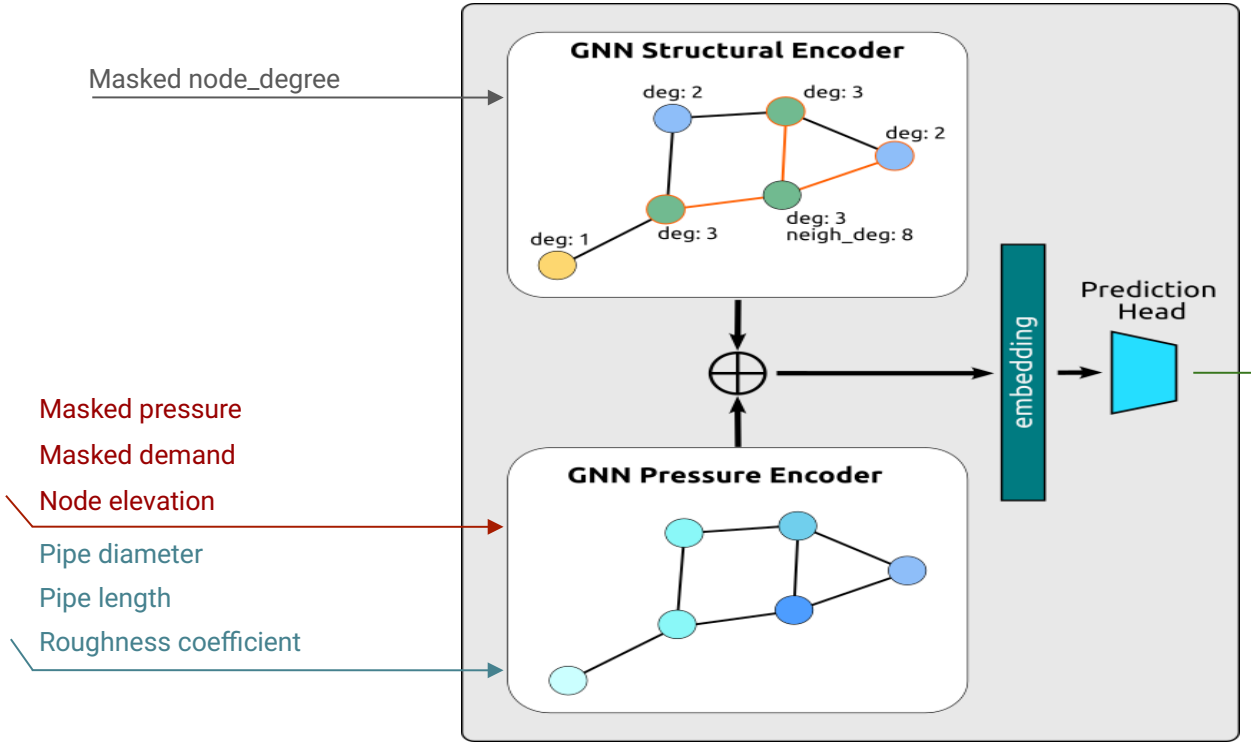
- **HuggingFace largest dataset for Graph Machine Learning**
  - 3.65 Terabytes of data
  - 1000 scenarios of WDN operation per network
  - More than 200 Million WDN states while operating under normal conditions.

- **Supporting Tasks:**
  - State Estimation
  - Demand Forecasting
  - Surrogate Modeling

Total downloads

165,653

Dual-Encoder



5-layer GATRes (Truong & Tello et al., 2024)

$$\hat{p} = \sigma(W_1 X_{\text{pressure}} + W_2 X_{\text{degree}})$$

$$loss = \lambda \times degree_loss \; + \; (1 - \lambda) \times pressure_loss$$

$$\lambda = \lambda_{min} + 0.5 \times (\lambda_{max} - \lambda_{min}) \times \left(1 + cos\left(\pi \frac{epoch}{total_epochs}\right)\right)$$

Model Evaluation on Unseen WDNs

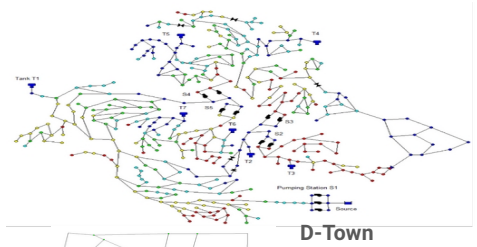
	Model	MAE	MAPE	R^2
D-Town	Baseline	1.9900 ±0.008	0.1020 ±0.002	0.9793 ±0.000
	DualEncoder (ours)	1.5642 ±0.005	0.0795 ±0.001	0.9857 ±0.000
	GFM fine-tuned (ours)	**1.3782 ±0.006**	**0.0705 ±0.001**	**0.9865 ±0.000**
Ky14	Baseline	1.8684 ±0.005	0.0516 ±0.000	0.9698 ±0.000
	DualEncoder (ours)	1.5846 ±0.003	0.0432 ±0.000	0.9764 ±0.000
	GFM fine-tuned (ours)	**1.5083 ±0.005**	**0.0407 ±0.000**	**0.9780 ±0.000**
WA1	Baseline	**0.0660 ±0.000**	**0.0093 ±0.000**	**0.9999 ±0.000**
	DualEncoder (ours)	0.3258 ±0.001	0.0275 ±0.000	0.9998 ±0.000
	GFM fine-tuned (ours)	0.3506 ±0.000	0.0313 ±0.000	0.9997 ±0.000
NPCL-1	Baseline	1.1893 ±0.002	0.0384 ±0.000	0.9916 ±0.000
	DualEncoder (ours)	1.0985 ±0.004	0.0372 ±0.000	0.9921 ±0.000
	GFM fine-tuned (ours)	**0.8841 ±0.002**	**0.0284 ±0.000**	**0.9945 ±0.000**
ky13	Baseline	1.6430 ±0.005	0.0289 ±0.000	0.9871 ±0.000
	DualEncoder (ours)	1.5639 ±0.004	0.0275 ±0.000	0.9892 ±0.000
	GFM fine-tuned (ours)	**1.3602 ±0.003**	**0.0238 ±0.000**	**0.9915 ±0.000**

Experiments

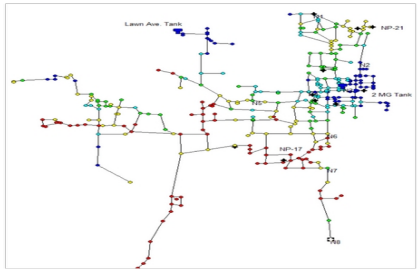
TRAINING

20 WDNs

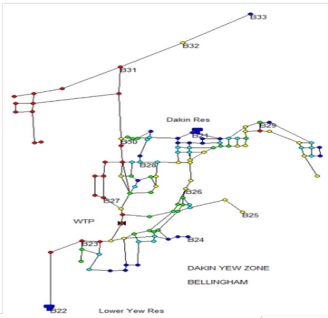
19 – 1900 nodes

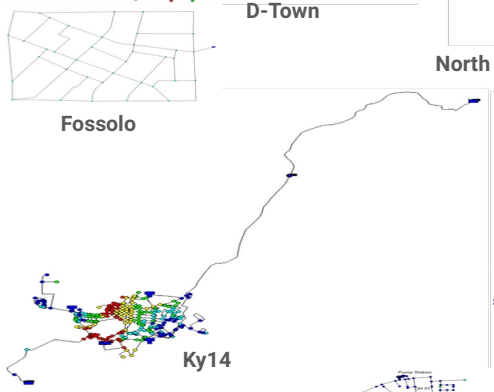


D-Town

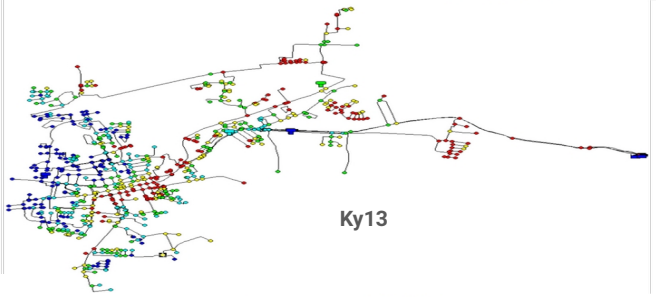


North Penn Water Authority System (NPCL-1)

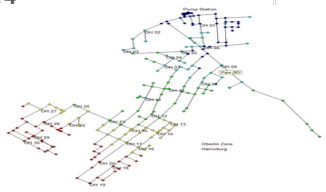




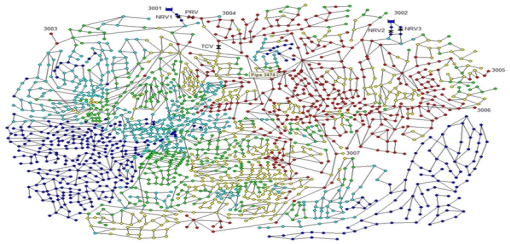
Ky14



Ky13



Oberlin (OBCL)



Exeter

TEST

8 Real WDNs



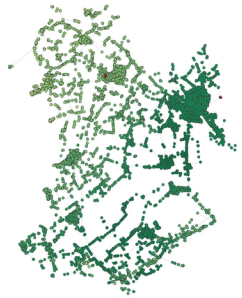
Almere



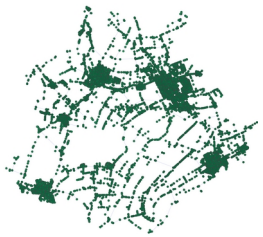
Zeewolde



Grou-Franeker



Sneek



Joure-Heerenveen



Gorredijk



Oosterwolde Drachten

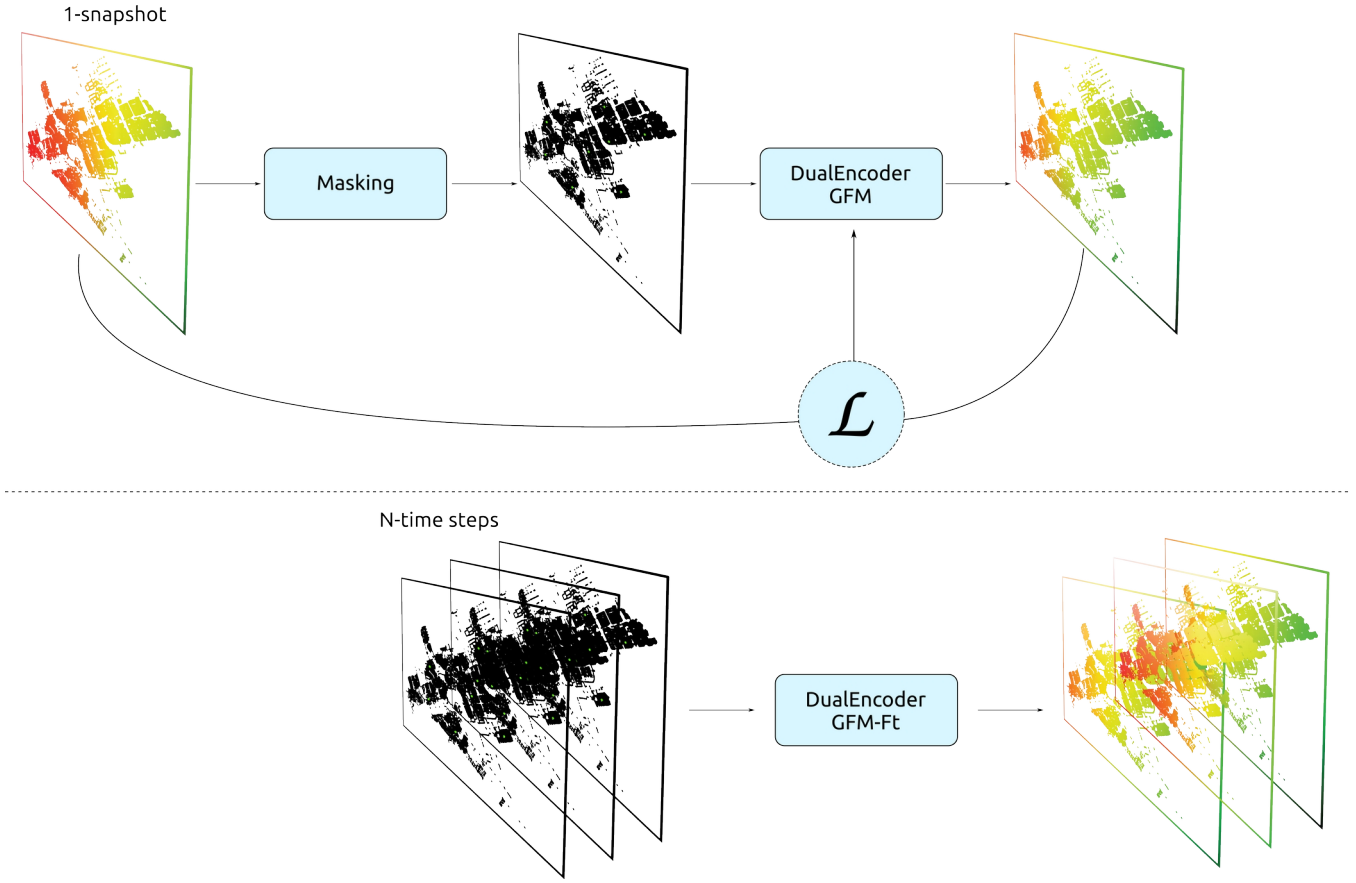


Noordwestelijk Weidegebied

5min: 288 snapshots 1tr / 1val / **286 test**

15min: 96 snapshots 1tr / 1val / **94 test**

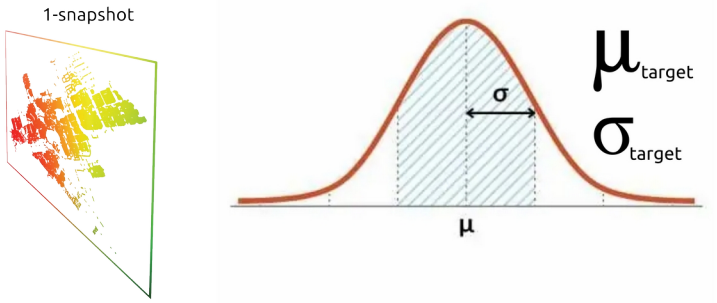
Fine tuning

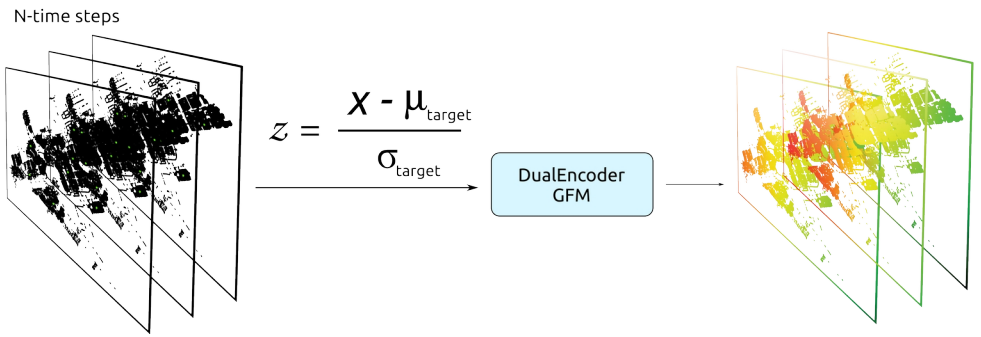


288 snapshots 1tr / 1val / **286 test**

96 snapshots 1tr / 1val / **94 test**

Target-aware normalization





288 snapshots: 1 stats, **287 test**

96 snapshots: 1 stats, **95 test**

Conclusions

- We presented GFM for WDNs, showing a better performance that single network training schemes.
- Pre-trained model on synthetic data + little adaptation on 1-snapshot perform very well on pressure recontruction.
- Feature aligment through 1-snapshot target-aware normalization replaces fine tuning in most cases.

Thank you

